

PRODIGY INFOTECH

Data Science Internship

Task-02: Data Cleaning and Exploratory Data Analysis (EDA)

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Internship: Data Science Internship

Task: 02

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

1. Objective

The objective of this project is to perform **Data Cleaning** and **Exploratory Data Analysis (EDA)** on a dataset to understand the relationships between variables, identify patterns, detect missing values and outliers, and generate meaningful insights through visualization.

2. Dataset

For this project, the **Titanic Dataset** was used.

The dataset contains passenger information such as:

- Passenger ID
- Survival Status
- Passenger Class

- Name
- Gender
- Age
- Fare
- Number of Siblings/Spouses
- Number of Parents/Children
- Embarked Port

This dataset is commonly used for data analysis and machine learning practice.

3. Libraries Used

Pandas

NumPy

Matplotlib

Seaborn

These libraries were used for data manipulation, statistical analysis, and visualization.

4. Data Cleaning

The following preprocessing steps were performed:

✓ Loaded the dataset

Imported the Titanic dataset into a Pandas DataFrame.

✓ Checked dataset information

Used:

`df.info()`

to understand column types and missing values.

✓ Checked missing values

`df.isnull().sum()`

Missing values were found mainly in:

- Age
- Cabin
- Embarked

✓ Handling Missing Values

- Missing Age values were replaced with the median age.
- Missing Embarked values were replaced with the mode.
- Cabin column contained many missing values and was removed.

✓ Duplicate Check

Verified that no duplicate records existed.

5. Exploratory Data Analysis (EDA)

Several visualizations were created to understand the data.

5.1 Dataset Overview

Displayed:

- First five rows

- Shape of dataset
 - Data types
 - Summary statistics
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5.2 Survival Count

A count plot was created to show:

- Survived
- Not Survived

Observation

More passengers did not survive than survived.

5.3 Gender Distribution

A count plot was used.

Observation

The dataset contains both male and female passengers, with males being more numerous.

5.4 Passenger Class Distribution

Passengers were grouped into:

- First Class
- Second Class
- Third Class

Observation

Most passengers belonged to Third Class.

5.5 Age Distribution

Histogram with KDE curve.

Observation

- Most passengers were between **20–40 years**.
 - Age distribution is approximately right-skewed.
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5.6 Fare Distribution

Histogram and Boxplot.

Observation

- Fare values are highly skewed.
 - Some passengers paid significantly higher fares.
 - Several outliers were identified.
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5.7 Survival by Gender

Bar chart comparing:

- Male survival
- Female survival

Observation

Female passengers had a much higher survival rate.

5.8 Survival by Passenger Class

Grouped bar chart.

Observation

Passengers travelling in First Class had the highest survival rate.

Third Class passengers experienced the lowest survival rate.

5.9 Correlation Heatmap

A heatmap was created using numerical features.

Observation

- Fare has a positive relationship with Passenger Class.
 - Age has only a weak relationship with survival.
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6. Key Insights

The analysis revealed several important findings:

- Female passengers were more likely to survive.
 - First Class passengers had better survival chances.
 - Third Class passengers experienced the highest mortality.
 - Younger and middle-aged passengers formed the majority of the dataset.
 - Fare distribution contains many outliers.
 - Missing values required preprocessing before analysis.
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7. Skills Learned

Through this project, the following skills were improved:

- Data Cleaning
- Handling Missing Values
- Exploratory Data Analysis
- Data Visualization
- Feature Understanding
- Statistical Summary
- Correlation Analysis
- Python Programming
- Pandas
- Seaborn
- Matplotlib

8. Conclusion

This project successfully demonstrated how **Exploratory Data Analysis (EDA)** can be used to understand a dataset before building machine learning models.

Data cleaning improved the quality of the dataset by handling missing values and removing unnecessary information.

Visualizations helped identify important relationships between variables, detect outliers, and understand passenger survival patterns.

Overall, this project strengthened practical skills in **Python**, **Pandas**, **Seaborn**, and **Matplotlib**, providing a strong

foundation for future machine learning and predictive analytics projects.